



- Domain: Travel & Tourism
- Data set: Flightbooking.csv



Flight Booking Price Prediction

Agenda

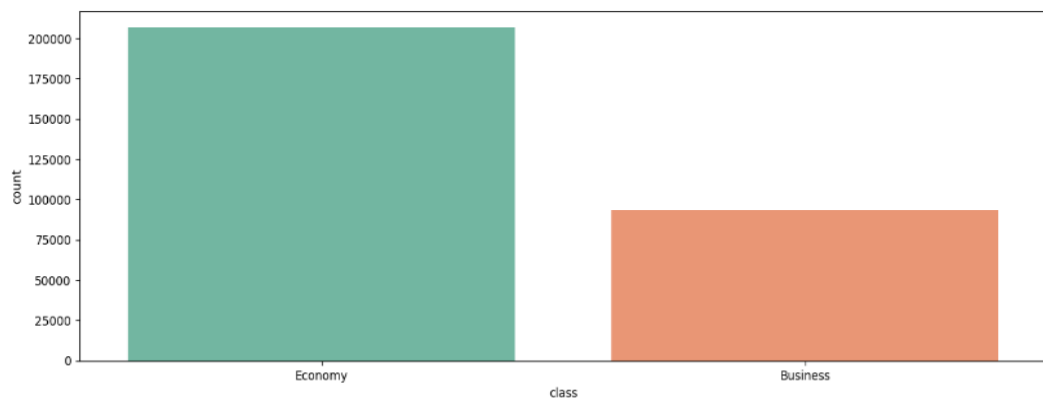
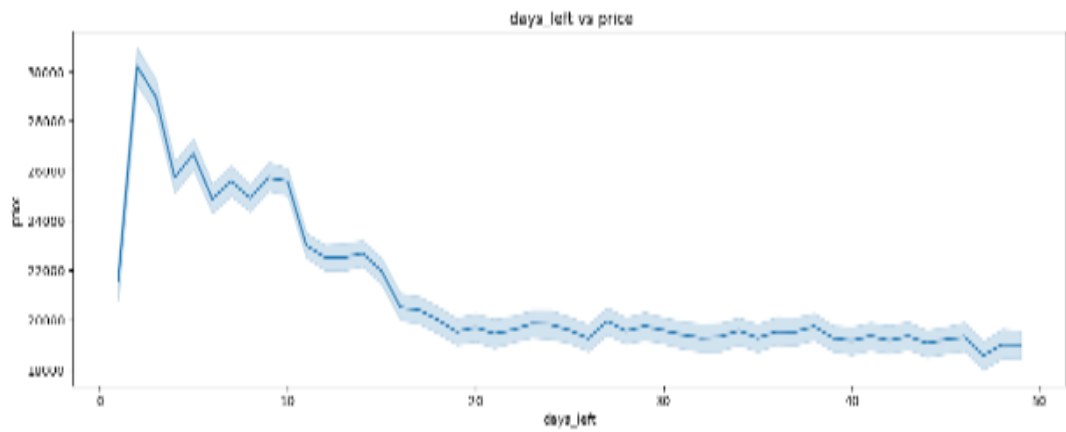
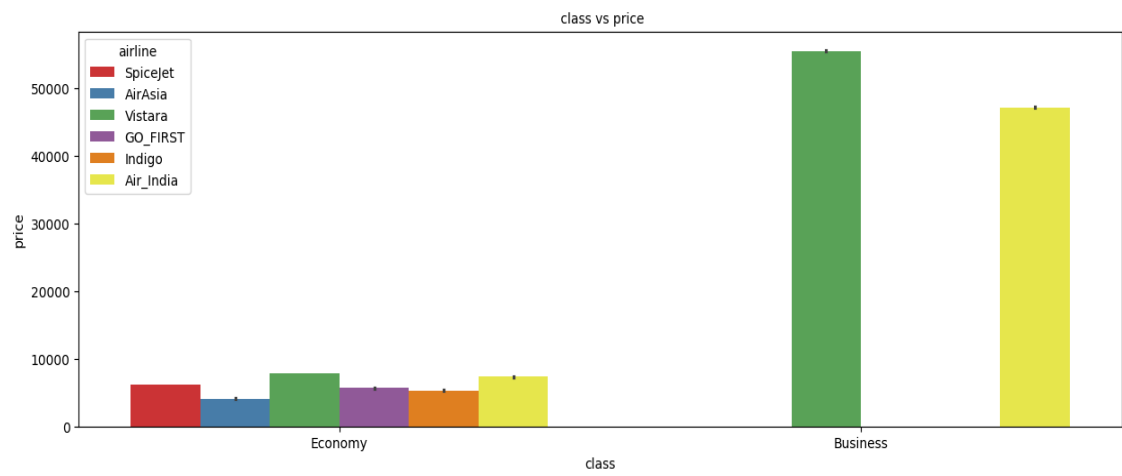
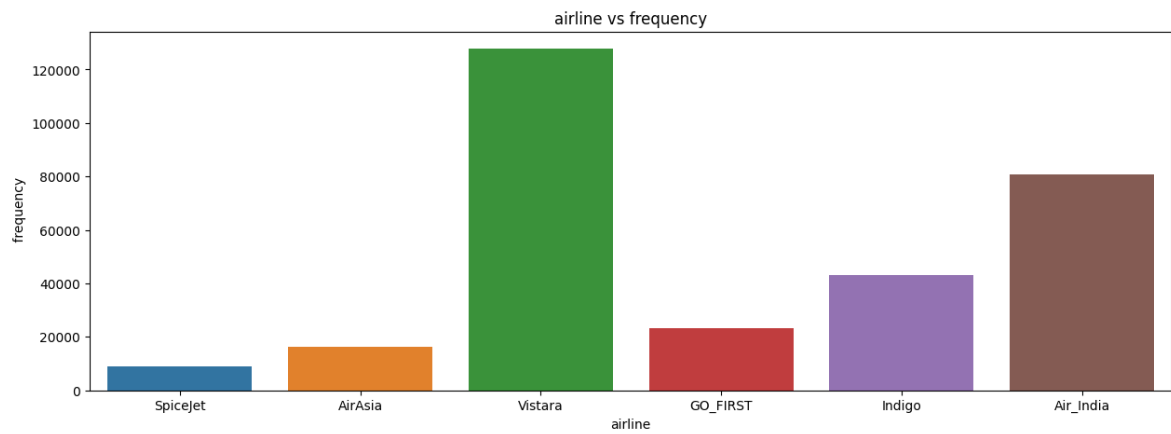
- 1 . Load the Libraries
2. Loading data
3. Data Visualization
4. One hot encoding
5. Features Selection
6. Implementing ML

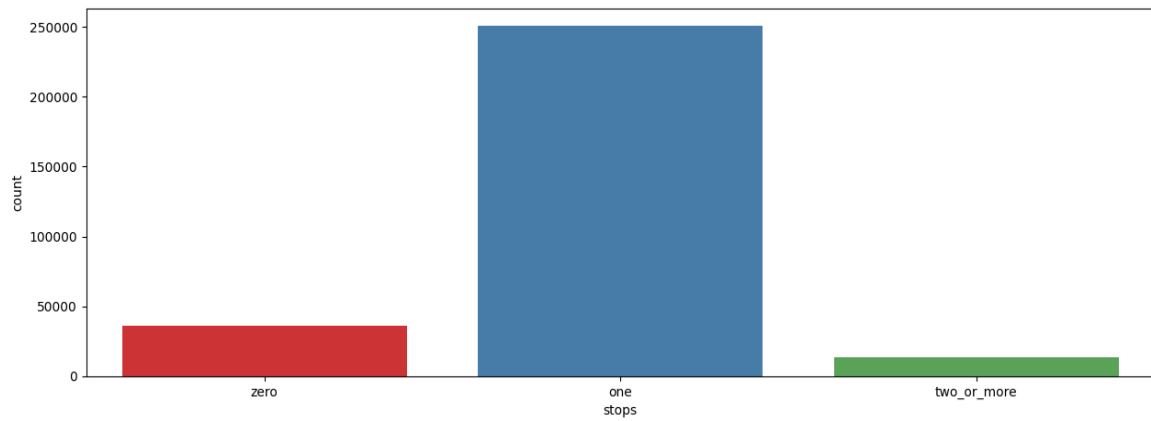
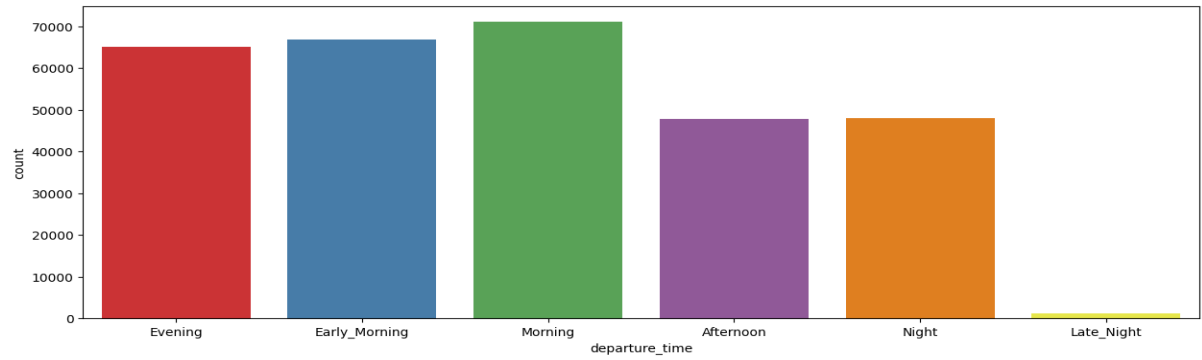
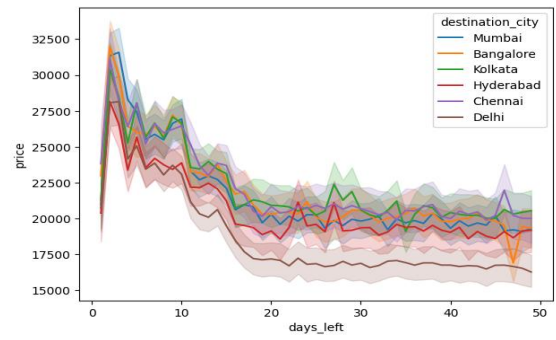
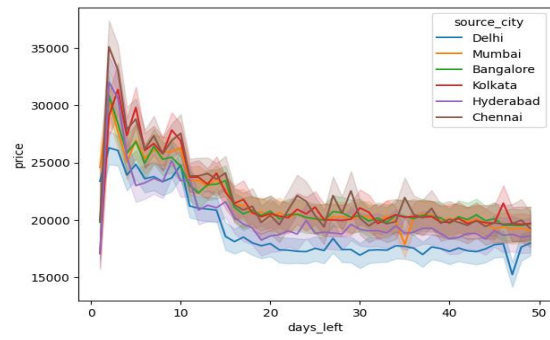
Problem Statement

The objective is to analyze the flight booking dataset obtained from a platform which is used to book flight tickets. A thorough study of the data will aid in the discovery of valuable insights that will enormous value to passengers. Apply EDA, statistical methods and machine learning algorithms in order to get meaningful information from it.

Attributes	Description
Airline	Name of the airline company
Flight	Plane's flight code
Source City	City from which the flight takes off
Departure Time	Time of Departure
Stops	Number of stops between the source and destination cities
Arrival Time	Time of Arrival
Destination City	City where the flight will land
Class	Contains information on seat class
Duration	Overall amount of time taken to travel between cities in hours.
Days left	Subtracting the trip date by the booking date.
Price	Ticket price

Exploring the features:

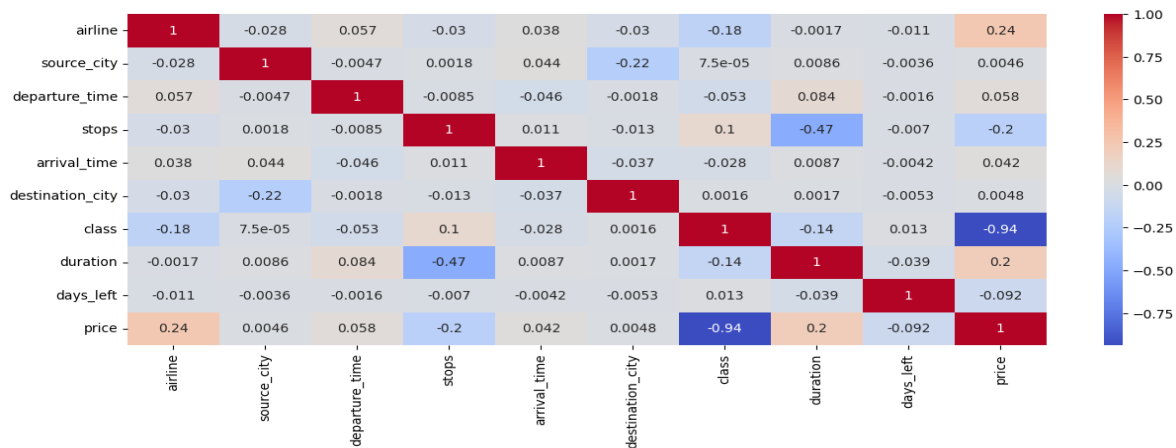




Feature Selection

`plt.figure(figsize=(15,5))`

`sns.heatmap(data.corr(),annot=True,cmap='coolwarm')`



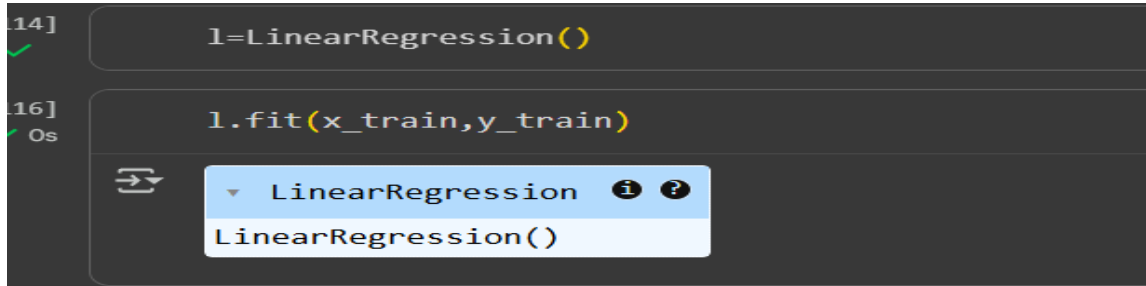
	features	VIF
0	airline	6.701818
1	flight	12.675046
2	source_city	2.969540
3	departure_time	2.806009
4	stops	1.437326
5	arrival_time	3.784873
6	destination_city	2.918792
7	class	2.852072
8	duration	4.754588
9	days_left	4.071591

	features	vif
0	airline	3.393124
1	source_city	2.927766
2	departure_time	2.779427
3	stops	1.426614
4	arrival_time	3.684550
5	destination_city	2.885337
6	class	2.849370
7	duration	4.113876
8	days_left	3.976790

ML MODEL

*from sklearn.linear_model import
LinearRegression*

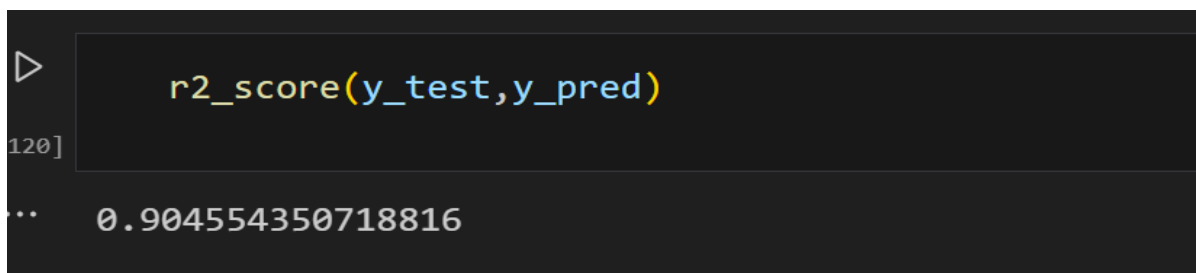
```
14] l=LinearRegression()  
16] l.fit(x_train,y_train)
```



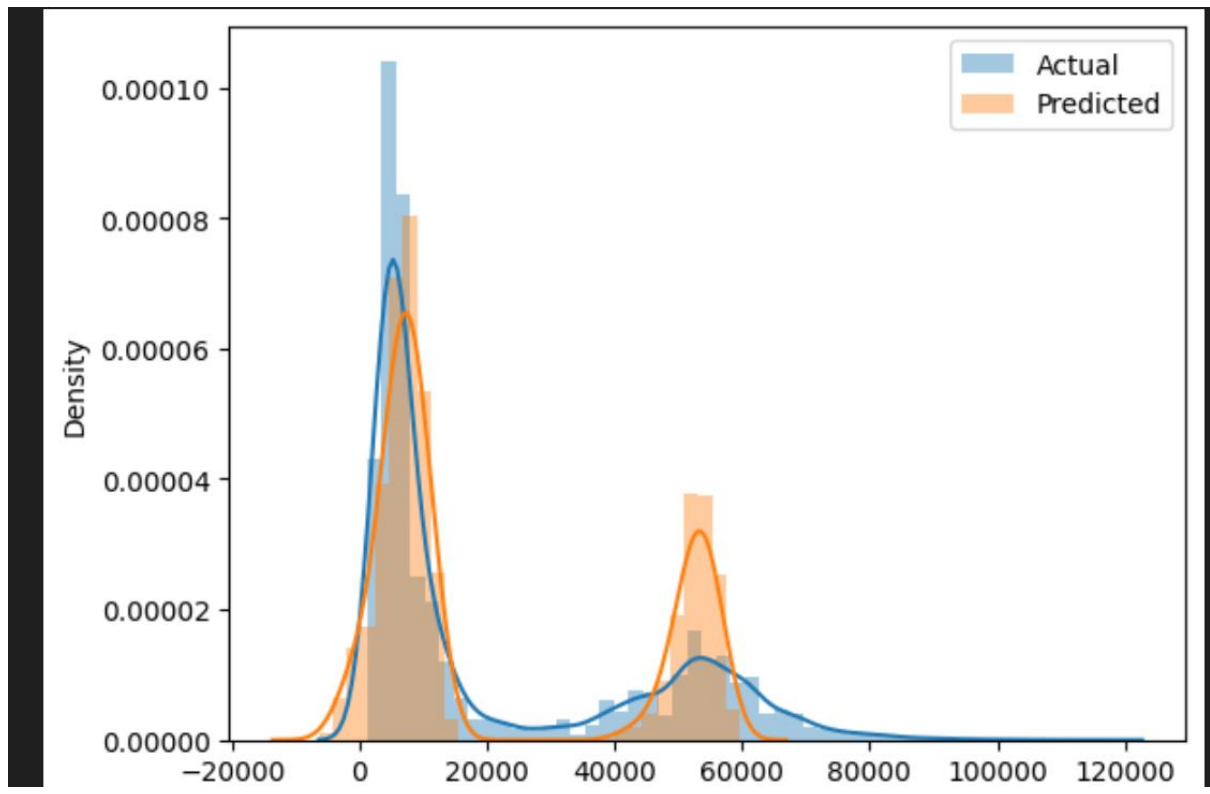
The image shows a Jupyter Notebook interface. On the left, there are two code cells with line numbers 14 and 16. Cell 14 contains the code `l=LinearRegression()` and cell 16 contains `l.fit(x_train,y_train)`. To the right of these cells, a dropdown menu is open, showing the variable `l` and its type `LinearRegression`. The dropdown menu has a blue header with a downward arrow, the text `LinearRegression`, and two icons (an information icon and a question mark icon). Below the header, the text `LinearRegression()` is visible.

	Actual_Value	Predicted_Value
0	7366.0	4585.880510
1	64831.0	52842.077073
2	6195.0	7877.980184
3	60160.0	56120.138579
4	6578.0	6663.204177
...	...	...
60026	5026.0	5829.230618
60027	3001.0	539.222230
60028	6734.0	5848.965500
60029	5082.0	-1498.061918
60030	66465.0	58762.406345

```
120] r2_score(y_test,y_pred)  
... 0.904554350718816
```



The image shows a Jupyter Notebook interface. On the left, there is a code cell with line number 120 containing the code `r2_score(y_test,y_pred)`. To the right of the code cell, the output of the function is displayed as `0.904554350718816`.



Data set

```
data=pd.read_csv("/content/drive/MyDrive/dmdw/Flight_Booking.csv")
data.head()
```

	Unnamed: 0	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price
0	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953
1	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953
2	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956
3	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955
4	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955

```
data.isnull().sum()
```

Unnamed: 0	0
airline	0
flight	0
source_city	0
departure_time	0
stops	0
arrival_time	0
destination_city	0
class	0
duration	0
days_left	0